

Use Case “Place Order”

1. Use case code

UC002

2. Brief Description

This use case describes how a Customer finalizes the purchase of items in their cart. It covers the entire process from validating the cart, providing shipping information, processing payment, and receiving final confirmation for the order.

3. Actors

3.1.Customer

4. Preconditions

- The Customer's cart contains at least one item.

5. Basic Flow of Events (Success)

1. The Customer requests to place an order from the cart screen.
2. The System checks if the stock quantity is sufficient for each product in the cart.
3. The System displays the delivery information form.
4. The Customer enters and submits their delivery information.
5. The System validates the delivery information.
6. The System calculates the shipping fee based on the total weight and delivery location, applying any applicable promotions
7. The System displays the final invoice screen, showing product prices, VAT, shipping fee, and the total amount.
8. The System calls the "Pay Order" use case.
9. After a successful return from the "Pay Order" use case, the System displays a confirmation screen with order details and transaction information. The screen presents two options: "Confirm Order" and "Cancel Order".
10. The Customer reviews the information and selects "Confirm Order".
11. The System saves the new order with a "pending" status and records the payment transaction.
12. The System sends a detailed order confirmation to the Customer's email.

13. The System displays a final "Order Successful" screen.

6. Alternative Flows

Table 1 — Alternative flows of events for UC "Place Order"

No	Location	Condition	Action	Resume location
1	At Step 2	The requested quantity for an item exceeds the available stock.	The System notifies the Customer which item is out of stock and displays the available quantity. The System prompts the Customer to update the cart.	Use case ends.
2	At Step 5	The delivery information is invalid (e.g., blank required fields, phone number format is incorrect).	The System displays an error message next to the invalid fields and prompts the Customer to correct them.	Step 4
3	At Step 8	The "Pay Order" use case returns unsuccessfully (payment failed).	The System displays a "Payment Failed" notification and allows the Customer to try again.	Step 7
4	At Step 10	The Customer selects "Cancel Order" on the final confirmation screen.	The System immediately processes a full refund using the original payment method, sends a cancellation confirmation email, and displays a cancellation success message.	Use case ends.

7. Input data

Table 2 — Input data for Delivery Information

No	Data fields	Description	Mandatory	Valid condition	Example
1	Receiver Name	Full name of the recipient	Yes	Not blank	Nguyen Van A
2	Phone Number	Contact phone number	Yes	10 digits, starts with '0', may contain '.', '-', ' '	0987-654-321
3	Province/City	Delivery location	Yes	Selected from a predefined list	Hanoi
4	Address	Street address details	Yes	Not blank	1 Dai Co Viet, Hai Ba Trung District
5	Shipping Instructions	Special notes for delivery	No	-	Please call before delivery.

8. Output data

Table 3 — Output data for Invoice Screen Data

No	Data fields	Description	Display format	Example
1	Product Title	Title of a media product	-	AIMS DVD
2	Price	Price per unit (including VAT)	Comma separator, VND	110,000 VND

3	Quantity	Quantity of the item	Integer	2
4	Amount	Total price for the item	Comma separator, VND	220,000 VND
5	Subtotal	Total amount for all products	Comma separator, VND	450,000 VND
6	Shipping Fee	Calculated delivery fee	Comma separator, VND	22,000 VND
7	Total	Final amount to be paid	Comma separator, VND	472,000 VND

Table 4: Successful Order Screen Data

No	Data fields	Description
1	Customer Name	Name of the person who ordered
2	Phone Number	Contact phone number
3	Shipping Address	Full delivery address
4	Total Amount	Total amount paid
5	Transaction ID	Unique ID from the payment gateway
6	Transaction Content	Brief description for the payment
7	Transaction Date	Datetime of the transaction

9. Postconditions

- On successful completion: A new order is created in the system with a "pending" status. A payment transaction is recorded. A confirmation email is sent to the Customer.
- On cancellation or failure: No order is created in the system. If payment was processed and then cancelled, a refund transaction is initiated.